

TALENTSIGNAL

The right people for any role — *by meaning, not keywords.*

Team EkaM · Sreenath M Menon

India.Runs × Redrob — Intelligent Candidate Discovery & Ranking



THE PROBLEM

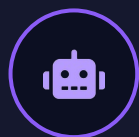
Recruiters miss the right person every day.

Not because the talent isn't there — because keyword filters can't see what matters.



The keyword blind spot

A great candidate who wrote “built the recommendation engine” scores zero against a JD that says “ranking systems.”



AI screeners fail differently

Newer LLM screeners confidently rank gamed résumés at the top — and can't explain or reproduce their own answer.

The cost: the best-fit candidate is invisible, and the ranking you get can't be trusted or defended.

OUR SOLUTION

One engine that ranks like a great recruiter would.



TalentSignal reads any job description, understands what the role truly needs, looks at the whole candidate — career, skills, behavioral signals — and returns a ranked, explainable shortlist you can trust.

By meaning, not keywords

No black box

No LLM in the ranking path

Same discipline a seasoned recruiter applies — made deterministic, auditable, and fast enough to run on 100,000 candidates.

WHY IT'S DIFFERENT

Three things keyword search and LLM screeners can't do.



Sees beyond keywords

Semantic + lexical matching over the JD's own requirements.

32

of our top 100 are people a keyword search would miss



Trustworthy & explainable

A verdict plus proof for every single candidate.

0

fabricated profiles reached our top 100



Production-real

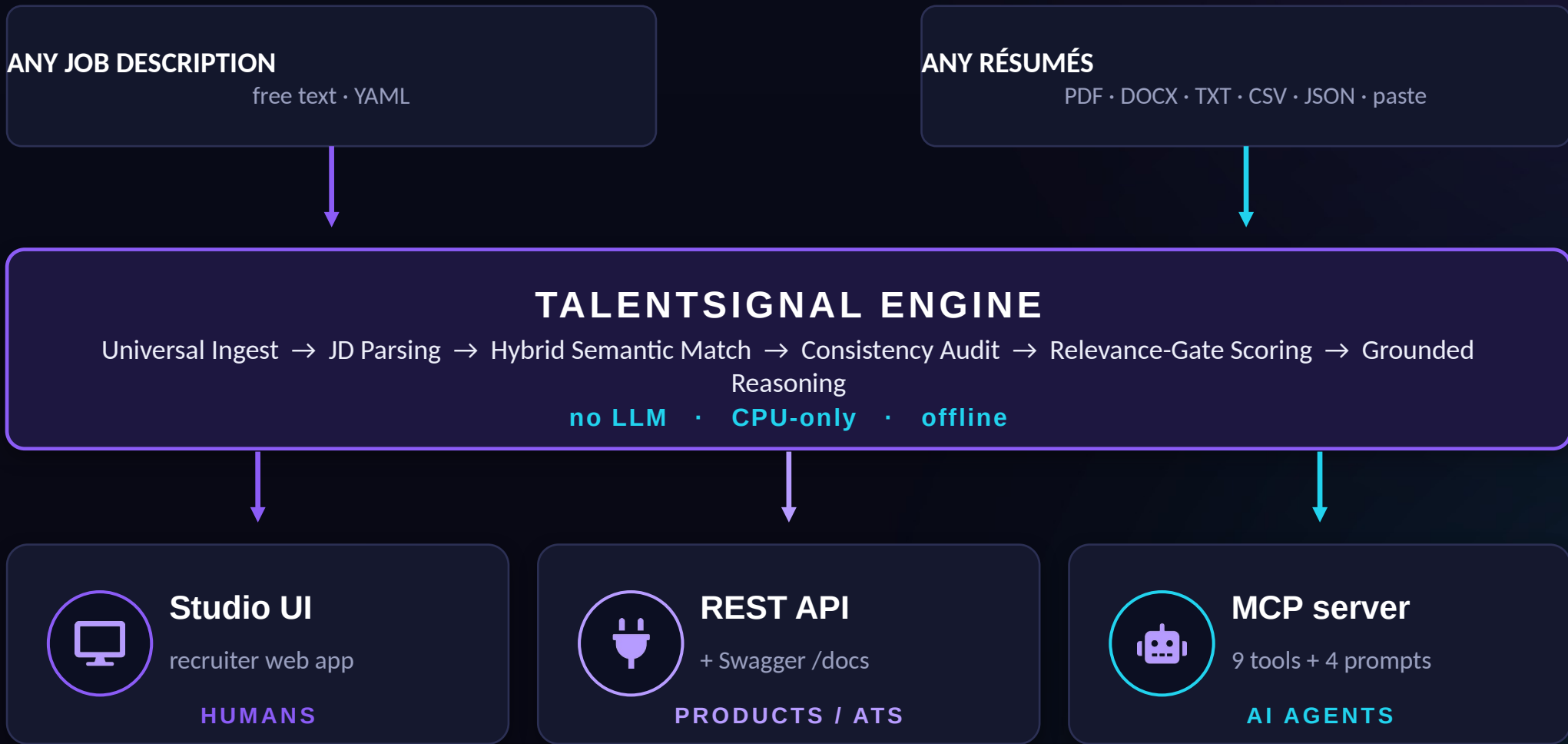
Deterministic, offline, CPU-only, no per-token cost.

<1 min

to rank 100,000 candidates end to end

SYSTEM ARCHITECTURE

One engine, three surfaces.



Same engine everywhere — a recruiter clicks it, a product embeds it, an AI agent calls it.

HOW RANKING WORKS

One law, driven entirely by the parsed JD.

$$\text{final} = R^{1.1} \times (0.45 + 0.55 \cdot Q) \times \text{soft_penalty} \times \text{hard_veto}$$

R

Relevance

Relevance to this JD's own requirements — the dominant term. Exponent 1.1 punishes weak relevance harder than it rewards.

Q

Quality

Seniority, logistics, behavioral, trust. Floored at 0.45 — it tunes the order, never overrides relevance.

S

Soft penalty

Graded risk and availability penalties — nudges, not knockouts.

V

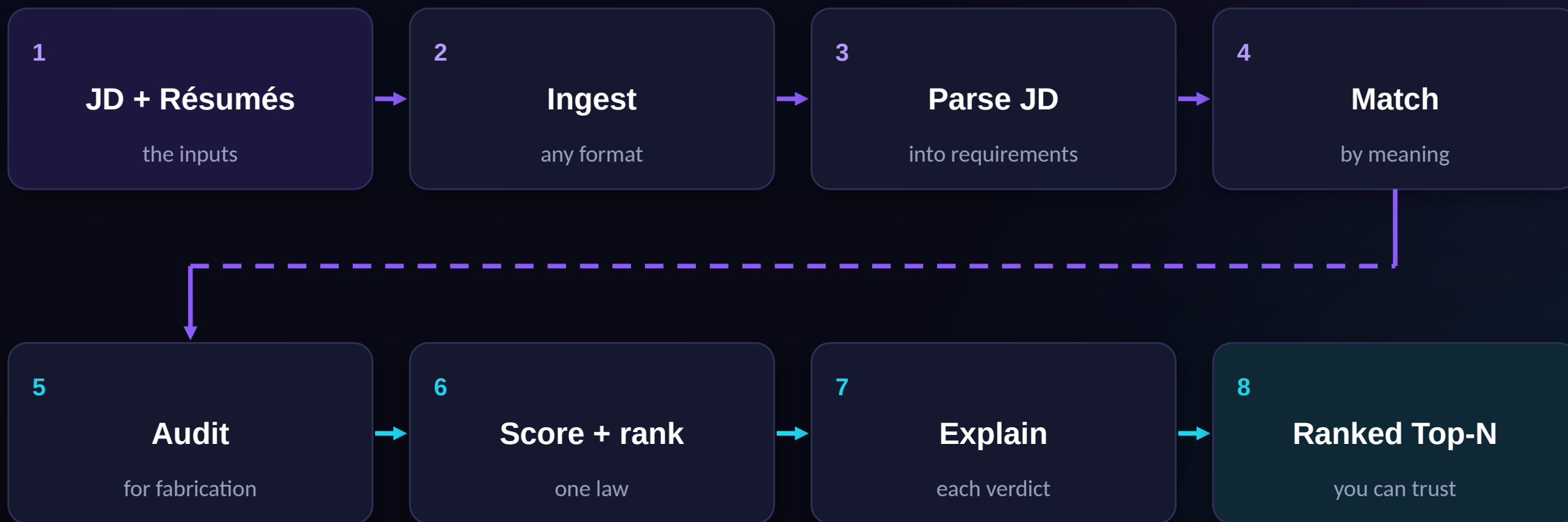
Hard veto

Zero only for an impossible profile or a true disqualifier with no counter-evidence.

No per-role code — the same law ranks an AI JD, a sales JD, or a designer JD.

MAIN WORKFLOW

From a job post to a shortlist you can trust — in one pass.



Same pipeline for the hackathon's 100,000 candidates and for a recruiter pasting five résumés.

THE SIGNATURE PROOF

The 32% a keyword search never sees.

KEYWORD SEARCH

Ranks by shared words.

"built the recommendation engine"
vs "ranking systems"

0 shared keywords → candidate missed



TALENTSIGNAL

Ranks by meaning.

"built the recommendation engine"
≡ "shipped a ranking system"

Matched with 0 shared words → surfaced

32

of our top 100 are invisible to a keyword filter.

On the real 100,000-candidate pool — rescued by meaning.

TRUST

It can't be fooled.



Role-independent consistency auditor

Flags impossible profiles by contradiction, not keywords:

- Expert skill claimed with 0 months of experience
- Tenure longer than the entire career

REAL EXAMPLE

A keyword-stuffed profile claiming 15 years at one company — but only 4 years total career.

✗ Flagged · score = 0

0

fabricated profiles reached top 100

100%

résumé-attack resistance (semantic)

78%

resistance for the keyword baseline

Tested across keyword-stuffing, prompt-injection, fabricated-experience and impossible-tenure attacks.

EXPLAINABILITY

Every decision, in plain sight.

For each candidate the recruiter sees four things — never a raw score.



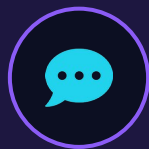
A verdict

Strong · Worth-a-look · Weak — the headline read at a glance.



Matched ✓ / Missing ✗

A skills panel scored against the JD's must-haves, not a generic list.



Grounded reasoning

1-2 sentences from the candidate's own words — audited so it never invents a skill or employer.



A reachability read

Reachable · passive · stale. Strong passive candidates are surfaced, never hidden.

The whole picture, not just the résumé.



Profile & career

Titles, years, and the career narrative — the story, not just a skills list.



Skills

Matched to the JD's requirements by meaning, weighted by what this role needs.



Behavioral signals

Activity, responsiveness, notice period, availability — used to re-rank, never to filter.

```
open_to_work  
  = false
```

“Not open” ≠ “won't join.” 64% of the real pool is not-open — and 22% of those are still reachable.
TalentSignal adapts to any signal vocabulary, not one dataset's fields.

Three surfaces, one engine.



Studio UI

recruiter web app · hosted

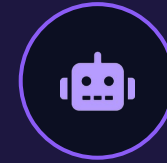
- Paste a JD + résumés → ranked, explained shortlist
- Compare candidates side by side
- Interview kits generated per candidate
- Live 100K “proof at scale” tab



REST API

OpenAPI + interactive Swagger

- /rank /ingest /audit
- /compliance /candidate_report
- Full OpenAPI spec
- Interactive Swagger at /docs



MCP server

the agentic surface

- 9 tools an AI agent calls directly
- 4 hiring-workflow prompts
- Works with Claude Desktop
- No chatbot — callable engine

WHY THIS MATTERS NOW

Built for when AI agents do the screening.

An autonomous AI recruiter needs an engine that is —



Callable

MCP tools, not a chatbot — an agent invokes it directly.



Explainable

Structured artifacts a human can audit and override.



Gaming-resistant

Resists résumé attacks that fool LLM screeners 30–95% of the time. We measured 100%.



Reproducible

Deterministic — the same answer replays for any dispute.

TalentSignal is the engine an AI recruiter can trust — *and a human can defend.*

RESULTS & PERFORMANCE

Proof, not adjectives.

32

rescued by meaning
in the top 100

0

fabricated profiles
in the top 100

100%

attack resistance
vs 78% keyword

0.95 / 0.97

ranking-quality
composite (labeled eval)

~30–67s

to rank 100K on CPU
budget was 5 min

234

automated tests
+ CI gate

Official validator: **PASS** — reproduces byte-identical output every run.

TECHNOLOGIES

Deliberately boring where it counts.

 **Python 3.11** the runtime

 **sentence-transformers** MiniLM · offline · CPU

 **NumPy** the only dep at rank time

 **stdlib HTTP** UI / API / MCP — no framework

 **OpenAPI + Swagger** typed, documented API

 **MCP** the agentic protocol

 **Git LFS** ships the precomputed index

 **234 tests + CI** the safety net



Key choice: no LLM and no GPU in the ranking path → fast, cheap, deterministic, and reproducible at 100K scale.

TALENTSIGNAL

The right people for any role — *by meaning.*



GitHub

`github.com/sreenathmmenon/talentsignal-ai`



Live Studio

`talentsignal-production.up.railway.app`



Live API + Swagger

`talentsignal-api-production.up.railway.app/docs`



Ranked output

`final_submission.xlsx` · `validator passes`